

Debugging Applications on Embedded Linux Targets

Removing bugs from software plays an important role in software development. Embedded Linux systems present their own challenges when they are being debugged. Learn how the GNU Project Debugger (GDB) addresses these issues.



Debugging is the process of finding and reducing the number of bugs in software. A debugger is the software tool that facilitates the debugging activity by the user in a methodical way. This process in the software life cycle gives an insight into how the software responds to different types of inputs given either by a user or by any external hardware or software.

Debug challenges on Linux systems

Debugging application code on embedded targets has unique challenges. The resources on the embedded target may simply be insufficient to support debugging. Smaller memory and non-volatile storage limitations can make it difficult to debug on the embedded target.

GDB (GNU Project Debugger) has been designed so that we can debug the application executing on the embedded target platform from the host PC. This provides an environment to use the resources of the host PC to debug the code executing on the embedded target. The host PC is usually a Linux box connected to the embedded target with an Ethernet cable.

User space and kernel space in Linux

In the Linux environment, the RAM is usually divided into two distinct regions—kernel space and user space. The kernel is the core of the operating system. It has the functionality of file management, process management, memory management and also the functionality to interact